UCS503 – Software Engineering Project

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ThaparTrade: Campus Marketplace for Student-to-Student Trading

A Structured Marketplace for Buying and Selling Products within the Thapar Community

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Chapter 1
Introduction

1.1 Background and Context

Students at Thapar frequently buy and sell textbooks, cycles, calculators, electronics, hostel items and other products within the campus community. These transactions are commonly handled through messaging groups, social media, personal contacts and word of mouth.

Such informal channels make listings difficult to discover and provide no structured mechanism for searching, filtering or tracking availability. ThaparTrade provides a centralized campus marketplace where authenticated students can create, discover and manage product listings.

1.1.1 Problem Statement

Existing student resale channels are fragmented and provide limited search, filtering and listing management. Buyers cannot easily find relevant products, while sellers lack a centralized platform for managing listings and availability.

ThaparTrade resolves this gap through a campus-focused marketplace where authenticated students can create, discover, manage and report product listings.

1.2 Project Objectives

The core objectives of the ThaparTrade project comprise:

1. **User Authentication:** Provide secure registration and login for students.
2. **Product Listing:** Allow users to create, edit and delete product listings.
3. **Marketplace Discovery:** Provide browsing, search and filtering of products.
4. **Listing Status:** Maintain Available, Reserved and Sold product states.
5. **Reporting and Moderation:** Allow users to report listings and administrators to review them.

Chapter 2
Methodology and Design

2.1 System Architecture

2.1.1 High-Level Design

ThaparTrade utilizes a modular three-tier architecture organized into the Client Marketplace Layer, the Application Layer, and the Database Layer. Figure 2.1 illustrates the high-level architecture.

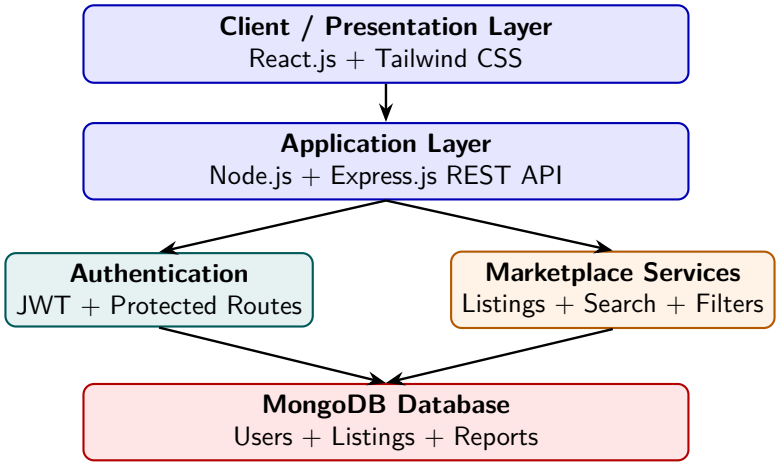


Figure 2.1: ThaparTrade High-Level Architecture

2.1.2 Technical Stack

The implementation stack was selected for a practical full-stack web application:

- **Frontend:** React.js and Tailwind CSS.
- **Backend Runtime & API:** Node.js and Express.js.
- **Database & Storage:** MongoDB.
- **Authentication:** JSON Web Tokens (JWT).
- **Image Storage:** AWS S3.
- **Payment Integration:** Razorpay.
- **Version Control:** GitHub.

2.2 Implementation Details

2.2.1 Module 1: Authentication & Product Listing

Users register and log in before accessing protected marketplace functionality. Authenticated users can create product listings with title, category, description, price, condition and images. Users can edit or delete their own listings and update product availability.

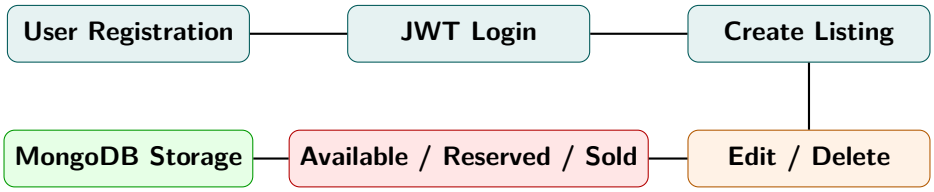


Figure 2.2: Authentication and Product Listing Flow

2.2.2 Module 2: Marketplace & Reporting

Users can browse, search and filter marketplace listings. Users can report problematic listings, after which an administrator can review the report and remove or restore the listing.



Figure 2.3: Marketplace Discovery and Reporting Flow

Chapter 3
Results and Evaluation

3.1 Testing Methodology

Testing followed a structured, multi-tier evaluation framework:

1. **Unit Testing:** Individual testing of authentication, listing and search functionality.
2. **Integration Testing:** End-to-end testing of REST APIs and MongoDB interactions.
3. **Scenario Validation:** Validation across representative buyer, seller and reporting workflows.

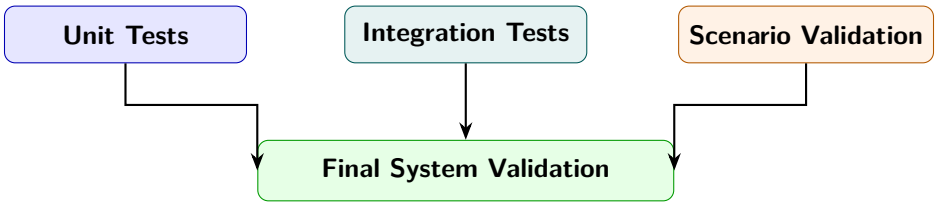


Figure 3.1: Multi-Tier Testing and Quality Assurance Framework

3.2 Performance Metrics

The prototype underwent functional evaluation to measure the reliability of its core marketplace workflows. Table 3.1 summarizes the evaluation metrics.

Evaluation Metric	Target Specification	Achieved	Status
Listing Creation	Functional	Implemented	Achieved
Search and Filtering	Functional	Implemented	Achieved
Listing Status	Available / Reserved / Sold	Implemented	Achieved
Reporting Workflow	Functional	Implemented	Achieved

Table 3.1: ThaparTrade Prototype Performance Metrics Summary

3.3 Analysis

The prototype successfully implements the core student marketplace workflow. Users can authenticate, create and manage product listings, browse products and use search and filtering functionality.

The listing status mechanism provides clearer availability information, while the reporting workflow provides a basic mechanism for administrative moderation.

Chapter 4
Conclusion

4.1 Summary of Work

During the first half of the project lifecycle, the team developed a functional full-stack ThaparTrade prototype. The system demonstrates user authentication, product listing management, marketplace discovery, listing status management and reporting.

4.2 Key Findings

1. **Centralized Marketplace:** A dedicated marketplace improves product discoverability.
2. **Structured Listings:** Category, price and condition provide structured product information.
3. **Status Management:** Available, Reserved and Sold states provide clearer availability.

4.3 Future Work and Recommendations

For the final project phase, the team will implement:

1. Improved search and filtering.
2. Enhanced buyer-seller communication.
3. Marketplace notifications.
4. Enhanced administrative moderation.

4.4 Final Remarks

ThaparTrade offers a practical campus marketplace for student-to-student trading. The prototype establishes the core workflow for authenticated users, product listings, marketplace discovery, availability management and reporting.